

Zihan(Lucy) Zhao

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EDUCATION

Northwestern University

Master of Science in Statistics and Data Science

Coursework: Causal Inference; Regression Analysis; Generative AI

University of International Business and Economics

Bachelor of Economics in Economic Statistics

Coursework: Time Series Analysis; Linear Regression; Statistical Inference; Econometrics

Evanston, IL

Sep 2024 – Jun 2026(Expected)

Beijing, China

Sep 2020 – Jul 2024

TECHNICAL SKILLS

Languages: Python, R, SQL, SAS, SPSS

Frameworks: Pytorch, Pandas, Numpy, Matplotlib, Spark(PySpark), Excel VBA

Analysis toolkits: Tableau, Power BI, Microsoft, AWS, Latex, Git, Google Suite

WORKING EXPERIENCE

Data Analyst Intern

Sina Weibo

Beijing, China

Mar 2024 – Jun 2024

- Improved the efficiency of **user feedback analysis** by 70% by developing and implementing a supervised NLP model, streamlining the process of identifying user needs.
- Developed group similarity metrics in Python and created a **Tableau dashboard** to visualize data, enabling stakeholders to make informed decisions on a weekly basis.
- Analyzed blog comment volume and interaction data via **SQL** to identify trending topics related to beauty products, informing co-branded content push strategies.
- Validated the strategy through **A/B testing**; Tracked KPIs such as click-through rate and comment volume, resulting in a 20% uplift across key user engagement metrics.

Financial Data Analyst Intern

Caida Securities Co Ltd

Beijing, China

Oct 2023 – Jan 2024

- Designed an **end-to-end** sentiment factor pipeline to analyze how financial news impacts short-term stock behavior across sectors, quantified through price movements.
- Built an **ETL pipeline** to extract and standardize **100K** records from 8 ChinaScope news datasets into 5 modeling-ready datasets with enriched sentiment and entity metadata.
- Benchmarked **TF-IDF** and **BERT** embeddings across **Lasso**, **Ridge**, and **Partial Linear Regression** models to capture the impact of news sentiment on near-term price behavior; selected **BERT + Lasso** for its 9.68% accuracy improvement and stronger interpretability.
- Used **Power BI** to visualize key predictive factors and the impact of news sentiment on short-term stock price movements; provided data-driven reports to support equity strategy team.

Data Consultant Intern

Ernst & Young

Beijing, China

Jul 2023 – Sep 2023

- Accessed client-side databases and used **PySpark** to clean, join and integrate multi-source financial and ESG datasets from insurance clients' portfolio companies.
- Created a hybrid **logistic regression** model with KMV features to improve prediction accuracy, and validated a performance gain of 8% using **ANOVA** testing.
- Built an **Excel VBA**-based portfolio construction tool that visualized risk-return tradeoffs and incorporated regulatory constraints, reducing manual work by **40%**.
- Used model-driven **credit risk** scores and **ESG** signals to guide insurance clients' portfolio decisions, contributing to **2+ green-transition investment proposals** targeting sustainable and low-risk firms.

PROJECTS

Health Chatbot

Evanston, IL

- Built a **RAG-based** clinical QA system using prompt-driven Gemini LLM and 3 chunking strategies over 200+ guideline segments, increasing answer accuracy from 45% to **71%** on 100 held-out patient queries.
- Fine-tuned a **LoRA-adapted** Gemini model on 300+ clinical QA pairs to enhance factual reasoning and reduce hallucination, improving response consistency by 24% based on manual evaluation.

LLM-enhanced Recommendation Engine

Evanston, IL

- Built a content-based recommender over 8 product features and enhanced it with **Gemini LLM** re-ranking to streamline results and diversify outputs; improved true user preference by **18.78%** in manual evaluation.
- Developed an **interactive interface** via **Streamlit** to visualize recommendation logic and diversity trade-offs; enabled users to toggle between "precise" and "diverse" modes, enhancing personalization and simplicity.